SDR_AVE
Advanced Data Average Analyzer for SDR#

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Introduction

SDR_AVE is a high-performance signal integration plugin for SDR#, fully ported to **.NET 9**. It features a slimmed-down, highly optimized codebase and a modernized graphical engine based on GDI+ with anti-aliased rendering.

The plugin is designed primarily for **radio astronomy**, weak-signal monitoring, and long-term spectral observations. Its core innovation lies in a two-stage linear integration engine combined with advanced interference mitigation and scientific-grade data export.

Installation: simply place SDRSharp.Average.dll in the Plugins folder — modern versions of SDR# will auto-detect and load the module.

Control Panel – Button & Function Guide

- **Enable Window** — activates IF spectrum processing and opens the visualization window
- **Gain / Level** — vertical scale and offset control (logarithmic view)
- **Signal Offset** — constant dB bias for power level calibration
- **Average** — number of frames used in the second-stage moving average (circular buffer size)
- **Background** — captures current average as noise reference
- **Reset Background** — clears stored background
- **Single / Multiple Save** — one-time or automated export of raw linear data
- **Color pickers** — persistent customization of text and graph fill colors
- **Median Filter** & **Savitzky-Golay** — toggles for RFI rejection and visual smoothing

Mathematical Engine – Two-Stage Linear Integration

Stage 1 – Intermediate Linear Integration

For each incoming block of IQ data of length N_{FFT}, the power spectrum of a single FFT frame is computed as:

$$\mathcal{P}_j[k] = \frac{|X_j[k]|^2}{N_{\text{FFT}}^2} = \frac{\text{Re}\big(X_j[k]\big)^2 + \text{Im}\big(X_j[k]\big)^2}{N_{\text{FFT}}^2}$$

where $X_j[k]$ is the k-th complex FFT bin of the j-th frame within the current intermediate block.

After that a median filter with (5-point Window) can be aplied as defined by the switch (on/of).

These M consecutive power spectra are then summed and averaged:

$$\begin{aligned} S[k] &= \sum_{j=1}^{M} \mathcal{P}_j\,[k] \\ P_i[k] &= \frac{S[k]}{M} = \frac{1}{M}\sum_{j=1}^{M} \mathcal{P}_j\,[k] \end{aligned}$$

where:

- M = Intermediate_average (user-configurable parameter)
- $P_i[k]$ — linear-averaged power for bin k after processing M FFT frames — this value is passed to Stage 2
- $k = 0,1,\dots,N_{\text{FFT}} - 1$

This first stage significantly reduces the data rate before long-term integration.

Square Root Transformation (Linearization): The normalized power data is linearized by calculating the square root of each frequency bin. This converts the signal from the power domain into the **Linear Magnitude (Amplitude)** domain. After this step, the data is stored in its strictly linear form.

Stage 2 – Finite Moving Average with Circular Buffer (True Arithmetic Mean)

Introduced in version 3.7 – replaces the previous infinite-memory EMA with a true moving average using a fixed-size circular buffer of length exactly N (value of the Average slider).

For each frequency bin k:

1. A new intermediate-averaged frame $P_i[k]$ arrives from Stage 1 (after optional median filter).
2. If the buffer is full ($i \geq N$), the oldest value $P_{i-N}[k]$ is subtracted from the running sum $S[k]$.
3. The new value $P_i[k]$ is added to $S[k]$.
4. The current moving average is computed as:

$$M[k] = \frac{S[k]}{\min(i,\, N)}$$

Once the buffer is full, the update rule becomes:

$$\begin{aligned} S[k] &\leftarrow S[k] - P_{i-N}[k] + P_i[k] \\ M[k] &= \frac{S[k]}{N} \end{aligned}$$

Key advantages over the previous EMA:

- Strictly finite memory: $\approx N \times N_{\mathrm{FFT}} \times 8$ bytes
- Equal weight $1/N$ for every frame in the window
- No long-term drift in noise floor during background subtraction
- Immediate clean reset when changing the Average value

Memory footprint example (maximum settings):

$$N = 50000,\ N_{\mathrm{FFT}} = 8192 \quad \Rightarrow \quad 50000 \times 8192 \times 8 \approx 3.3\ \mathrm{GB}$$

Typical use ($N = 500 - 5000$) → 30–300 MB.

Total Integration Length

The total number of original FFT frames contributing to the currently displayed spectrum is:

$$\text{Total frames} = M \times \min(N,\ \text{number of processed intermediate blocks})$$

After the circular buffer is full, this value stabilizes at:

$$M \times N = \text{Intermediate\_average} \times \text{Average}$$

Background Subtraction & Bias Correction

When background recording is enabled, the current moving average $M[k]$ is stored as the reference buffer $B[k]$.

Subsequent spectra are corrected in the logarithmic domain:

$$\mathrm{dB}_{\mathrm{final}}[k] = 10\log_{10}(M[k] + \varepsilon) - 10\log_{10}(B[k] + \varepsilon)$$

where $\varepsilon = 10^{-20}$ (avoids log(0)), and Δ_{bias} (default $+0.1\,\mathrm{dB}$) compensates for small numerical/statistical offsets.

Why it matters for Long-term Observation: When you record a "Background" reference, you are capturing a static snapshot of the noise floor. Because the **Circular Buffer** ensures that every subsequent "Live" frame is calculated using the exact same number of samples (N) as the background, the subtraction (**Live - Background**) remains mathematically consistent even after days of continuous operation.

Advanced Interference Mitigation: Median Filter (RFI Rejection)

A 5-point median filter is applied independently per frequency bin before Stage 2:

$$P_{\mathrm{med}}[k] = \mathrm{median}(P_{i-2}[k],\ P_{i-1}[k],\ P_i[k],\ P_{i+1}[k],\ P_{i+2}[k])$$

A frame is flagged as rejected if its power at the center bin exceeds the median by more than 10 dB:

$$P_i[N_{\mathrm{FFT}}/2] > 10 \times P_{\mathrm{med}}[N_{\mathrm{FFT}}/2]$$

The rejection counter is displayed live in the visualization window.

Visual Enhancement: Savitzky-Golay Smoothing

Applied **only to the displayed data** (after background subtraction):

$$y_i = \sum_{j=-2}^{2} c_j\ d_{i+j}, \qquad \mathbf{c} = \left[-\frac{3}{35}, \frac{12}{35}, \frac{17}{35}, \frac{12}{35}, -\frac{3}{35}\right]$$

Important: exported files always contain raw linear values $M[k]$ — no Savitzky-Golay smoothing is ever applied to saved data.

Data Export & Scientific Integrity

Exported text files contain raw linear power after Stage 2:

```
Frequency (MHz)  Linear Magnitude
86.126000000     0.000000210
86.135765625     0.000000228
...
```

This format preserves full statistical independence between bins and enables accurate energy calculations in external tools (Python, MATLAB, etc.).

Automated Acquisition Sequence: Multiple Save Mode

In version 4.5, the **Multiple Save** functionality is governed by a precise state-machine to ensure data purity across periodic observations. Each recording cycle follows this rigorous operational flow:

1. Cycle Initialization (The Reset Stage)

- Upon starting a new record, the plugin automatically triggers a **Buffer Reset.**
- The **Circular Buffer** and all energy accumulators are purged.
- The `CurrentSampleCount` is zeroed out to ensure that the new file contains no "leakage" or "spectral ghosting" from the previous observation.
- **Note:** The recorded **Background** reference remains persistent and is NOT reset, maintaining consistent calibration across the entire multi-save session.

2. Deterministic Acquisition

- The engine begins collecting new FFT frames. Each frame passes through the **5-point Median Filter** (if enabled) and preintermediate average. After that is linearized via the **Square Root** of the power data.
- The data fills the circular buffer until the exact number of frames defined by the *Average* slider (target count) is reached.

3. Data Export (The Save Stage)

- Once the buffer is full, the integrated **Linear Magnitude** data is calculated.
- The result is exported to a timestamped text file.
- **Reminder:** Because the data is stored in its linear amplitude form, external analysis (Python/MATLAB) must apply the **20 log10** coefficient to convert these values back to Decibels (dB).

4. Inter-Record Delay

- The system enters a "Wait" state for the duration specified in the **Delay** setting.
- During this interval, processing is suspended to reduce CPU load and maintain system responsiveness.

5. Automated Re-Loop

- Once the delay timer expires, the sequence automatically returns to **Step 1 (Reset)**.
- A new independent measurement begins, ensuring each file in the series represents a unique, isolated window of time.

Practical Recommendations & Best Practices

- Enable Median Filter in urban or RFI-heavy environments
- Set Intermediate_average = 1–5 for optimal impulse detection
- Use Average = 500–5000 for a good balance between SNR gain and responsiveness
- Enable Savitzky-Golay when searching for narrow weak lines (HI, OH, pulsars)
- Record background for at least 2–3× the planned observation duration
- Export data frequently – linear format is irreplaceable for scientific post-processing